Abnormal Uterine Bleeding and Pelvic Pain

DANIELLE CULLEN, MD

(DANIELLE.CULLEN@JEFFERSON.EDU)

CLINICAL ASSISTANT PROFESSOR

OBSTETRICS AND GYNECOLOGY

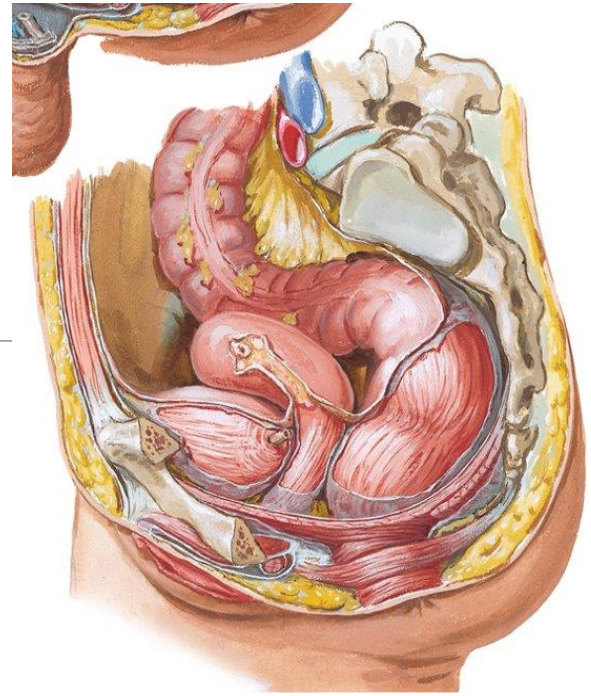
OCTOBER 29, 2026

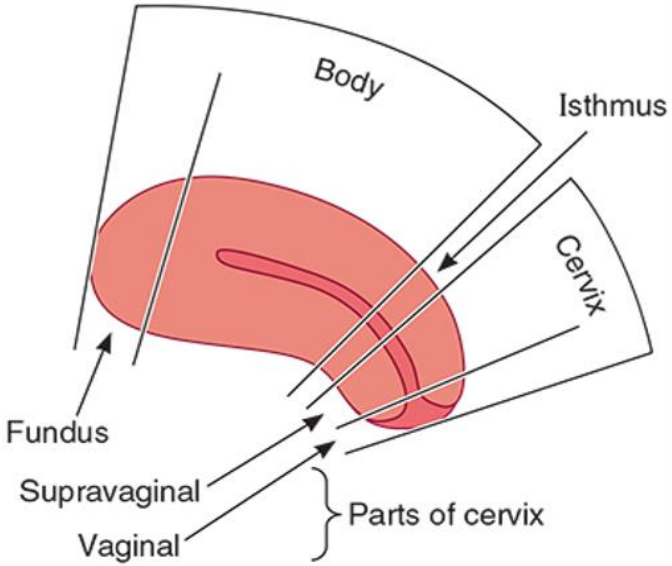
Learning Objectives

1. Describe normal uterine anatomy
2. Explain normal uterine bleeding patterns as a framework for recognizing abnormal uterine bleeding patterns
3. Discuss causes of abnormal uterine bleeding (AUB) in **reproductive age** people (PALM-COEIN), including clinical presentations, diagnosis and treatment options for each
4. Recognize other (non-uterine) causes of vaginal bleeding
5. Identify common causes of postmenopausal bleeding
6. Describe the imaging modalities commonly used for diagnosing cause of AUB
7. Recognize and describe the common causes of chronic pelvic pain

Normal Anatomy of the Uterus

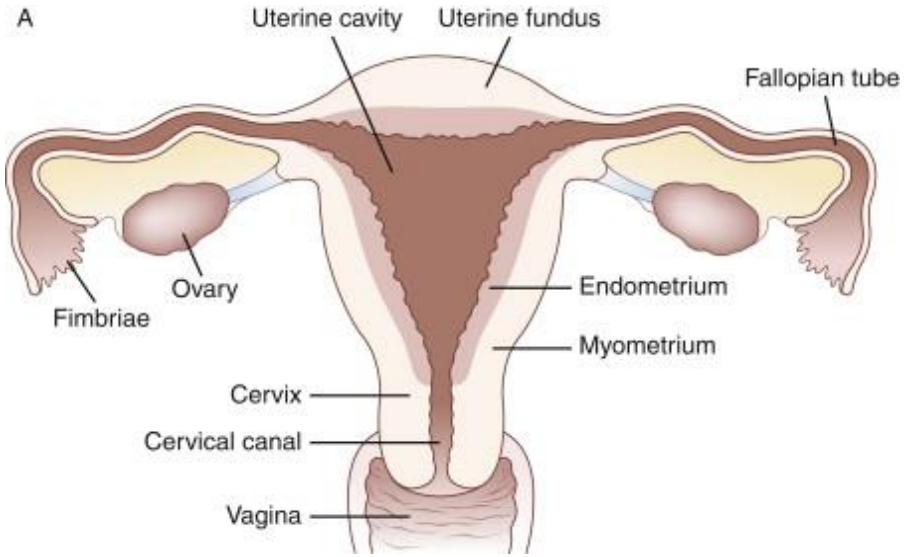
- Thick-walled, hollow, muscular organ
- Potential space inside the uterus ("endometrial cavity")
- Normal nonpregnant uterus lies within the lesser pelvis
  - Two parts: body and cervix
  - Body lies on the roof of the urinary bladder and the cervix placed between the urinary bladder and rectum

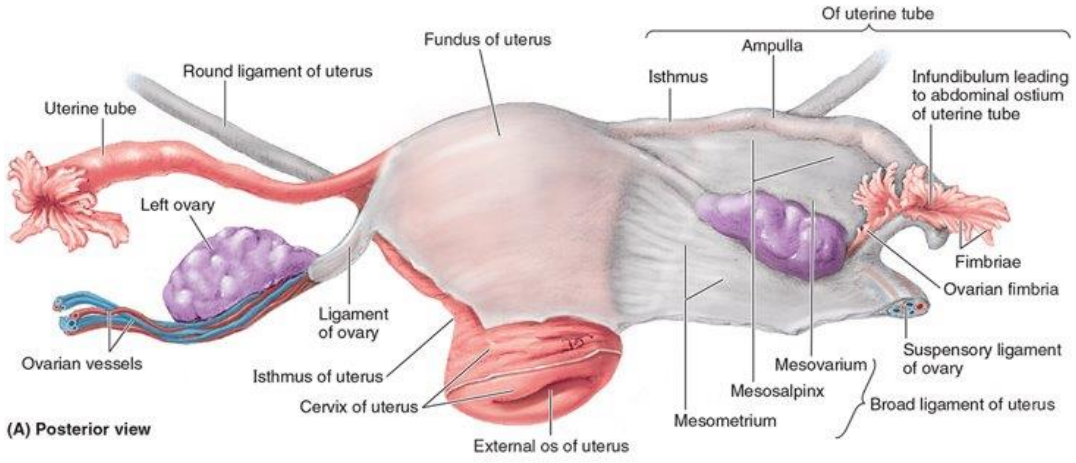




Normal Anatomy of the Uterus

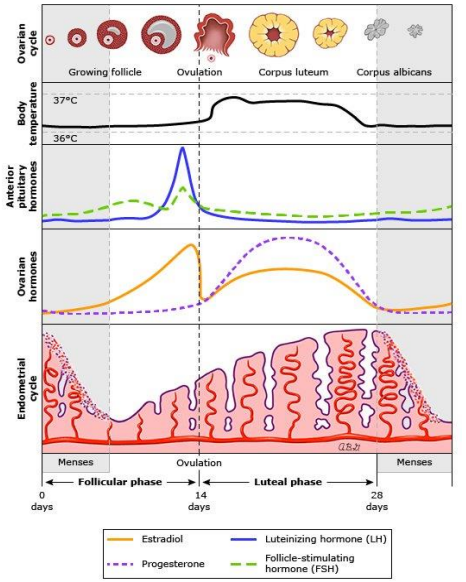
- Three layers
  - Perimetrium- external (serosal layer, peritoneum)
  - Myometrium- middle layer (smooth muscle layer)
  - Endometrium- internal layer
- "-metrium" refers to the layers of the uterus
- Laterally the perimetrium becomes the broad ligament

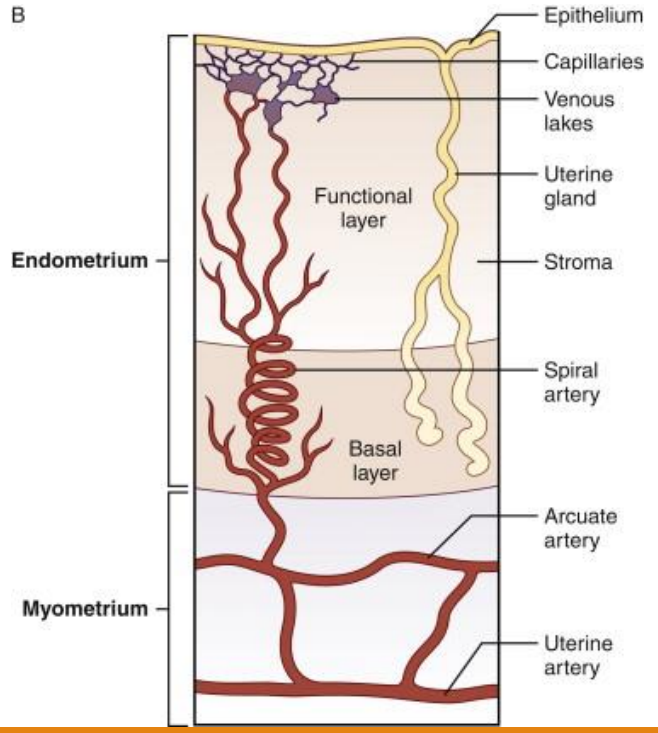




Normal anatomy of the Uterus

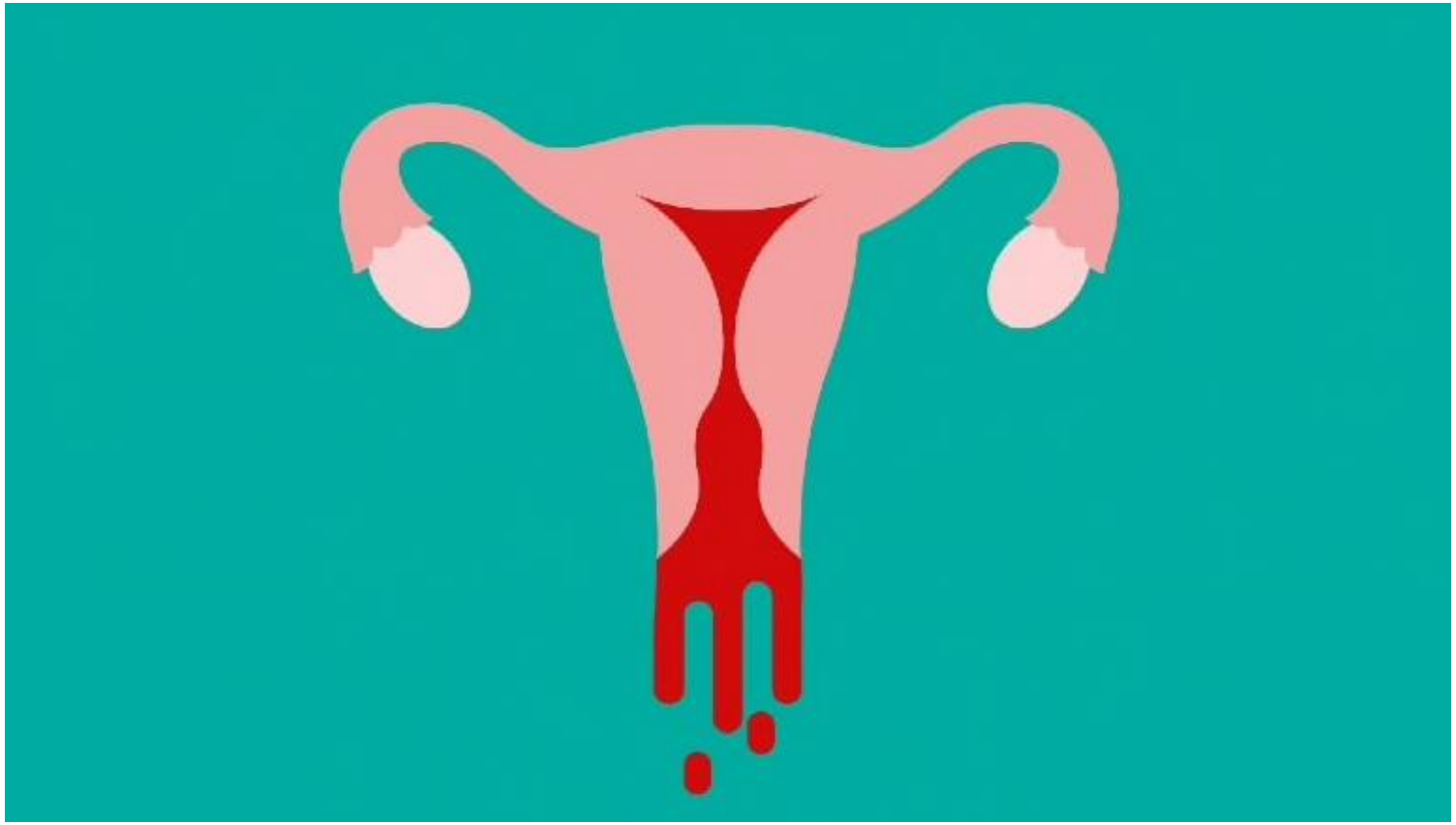
- Endometrium- internal to the myometrium, lines the uterine cavity
- Divided into basal layer (*stratum basalis*) and functional layer (*stratum functionalis*)
  - Functional layer: lining layer of epithelium, venous lakes, capillaries, uterine glands and stroma- **shed during menses**
    - **Uterine glands and stroma elsewhere is endometriosis**
    - **Uterine glands and stroma in the myometrial layer is adenomyosis**
  - Basal layer does not shed during menstruation, but its vessels can contribute to normal or abnormal menstrual flow
    - After menses, during the proliferative phase, the basal layer gives rise to a new functional layer- hormonally regulated!

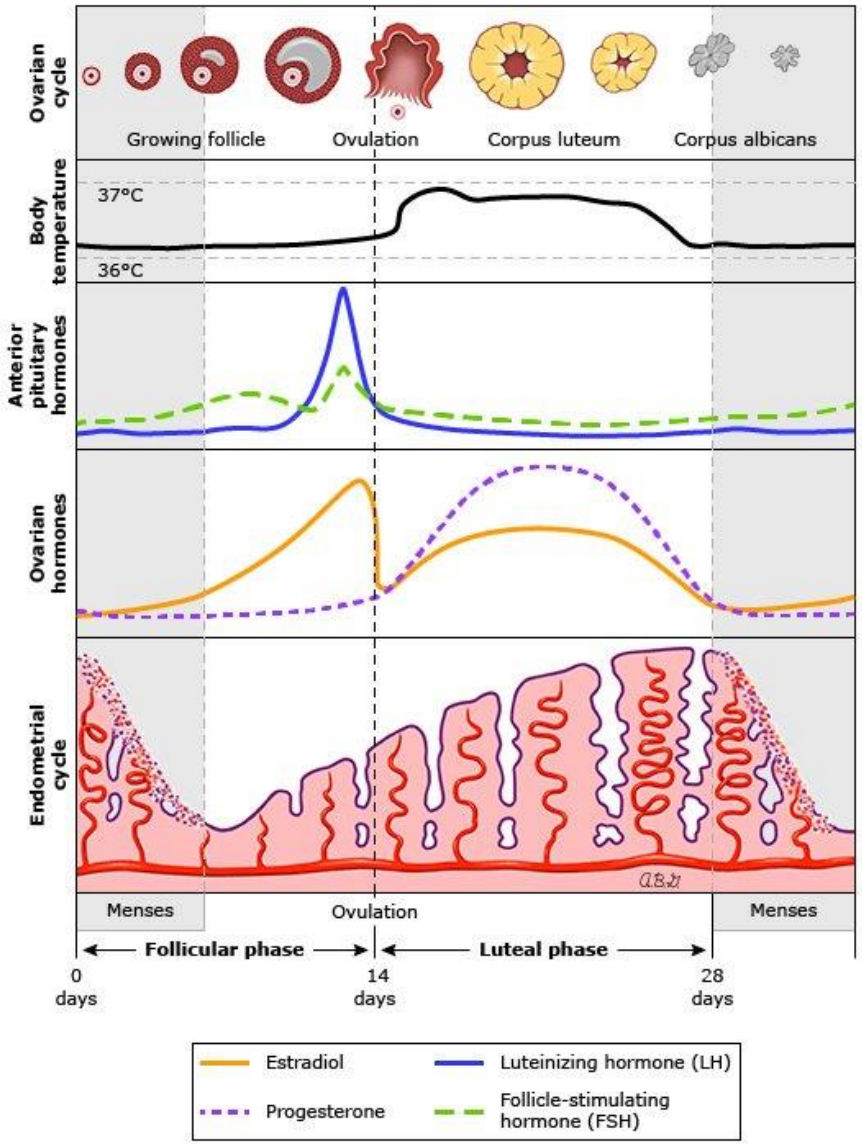




WHY IS AUB IMPORTANT?

AUB=
Abnormal
Uterine
Bleeding





Review of Normal Uterine Bleeding

Framework for abnormal uterine bleeding

Normal menstrual cycle 24-38 days

- Follicular phase (includes menses)
- Luteal phase

Duration of menses $\leq$ 8 days

Generally regular with variation $\leq$7-9 days

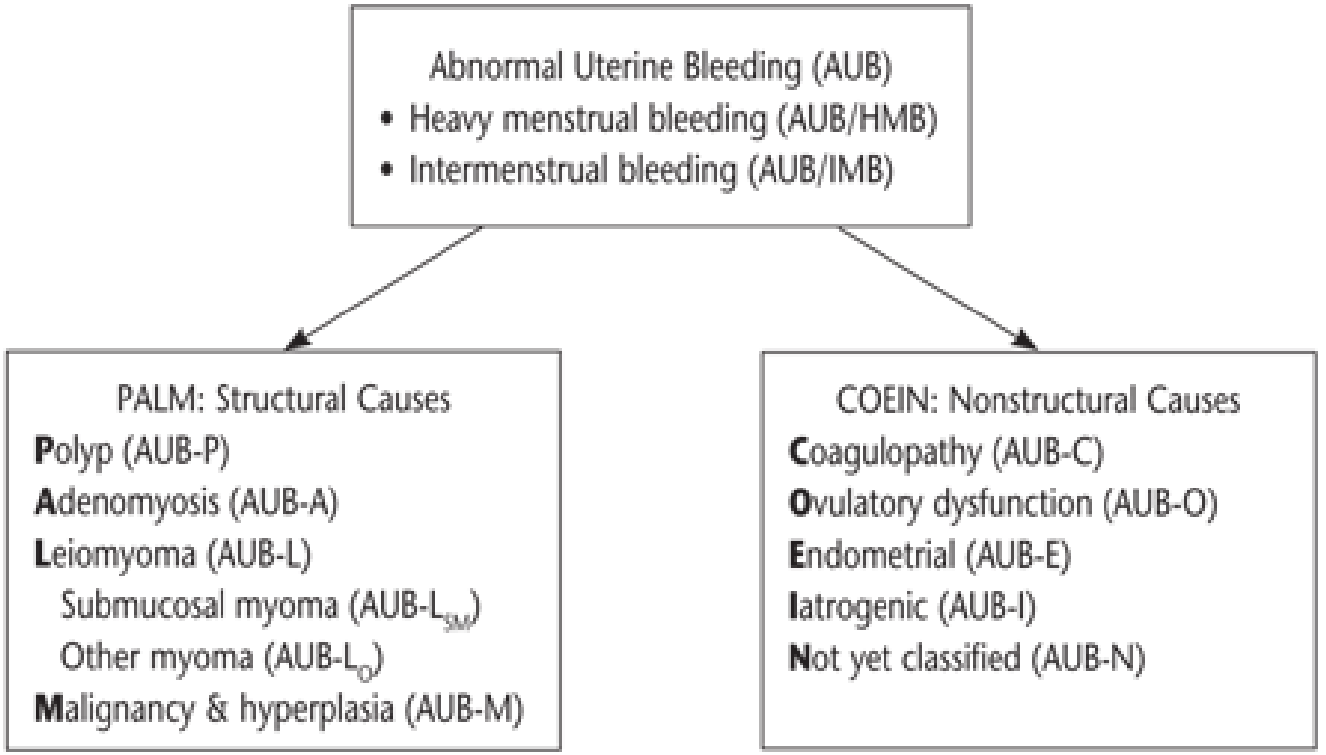
FIGO Definitions and Terminologies

Parameter	Normal	Abnormal
Frequency	Absent (no bleeding) = amenorrhea	
	Infrequent (>38 days)	
	Normal (≥24 to ≤38 days)	
	Frequent (<24 days)	
Duration	Normal (≤8 days)	
	Prolonged (>8 days)	
Regularity	Normal or "Regular" (shortest to longest cycle variation: ≤7-9 days)*	
	Irregular (shortest to longest cycle variation: ≥8-10 days)*	
Flow Volume (patient determined)	Light	
	Normal	
	Heavy Heavy Menstrual Bleeding = "HMB"	

Intermenstrual Bleeding (IMB) Bleeding between cyclically regular onset of menses	None	
	Random	
	Cyclic (Predictable)	Early Cycle
		Mid Cycle
		Late Cycle

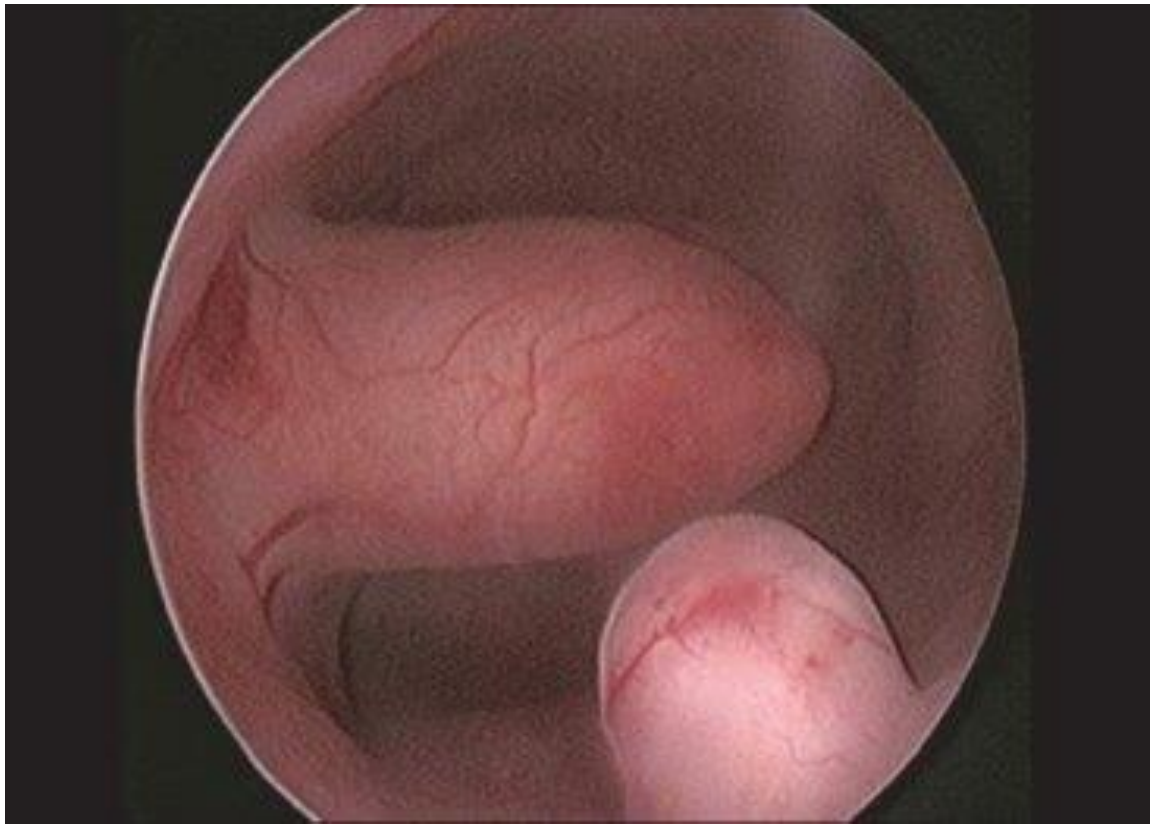
Causes of Abnormal Uterine Bleeding in **Reproductive-Age** People

PALM-COEIN Classification



Structural Causes (PALM)

PALM-COEIN



Endometrial Polyp

Hyperplastic overgrowth of endometrial glands and stroma

Project from surface of the endometrium

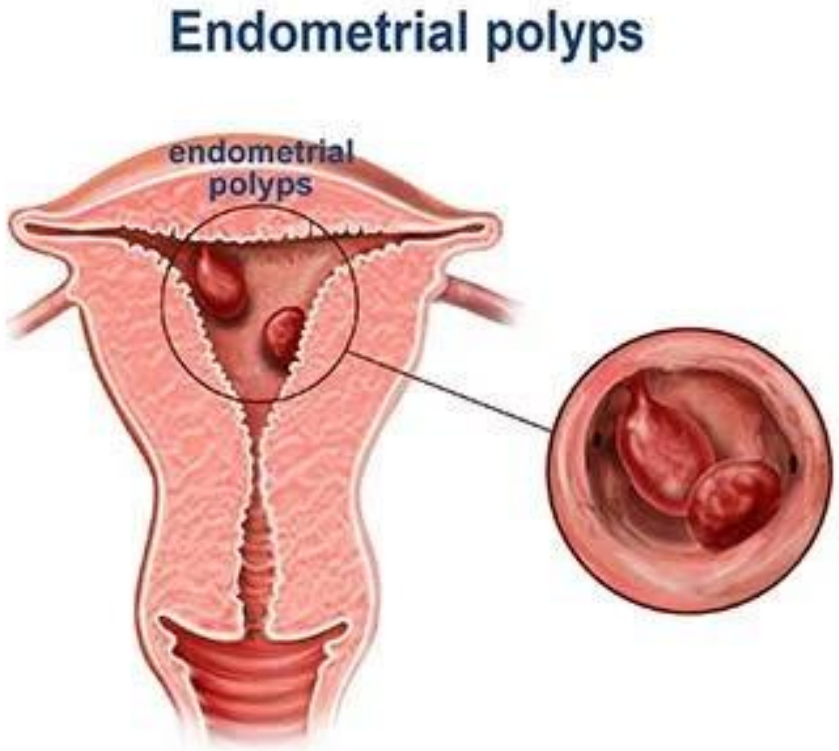
Most often benign, malignancy can occur

Seen in both pre and postmenopausal people

- More common between 40-50 yo

Endometrial Polyp: Clinical Presentation

- Intermenstrual bleeding
- Heavy and/or prolonged menses
- Postcoital bleeding
- Asymptomatic
- Infertility?

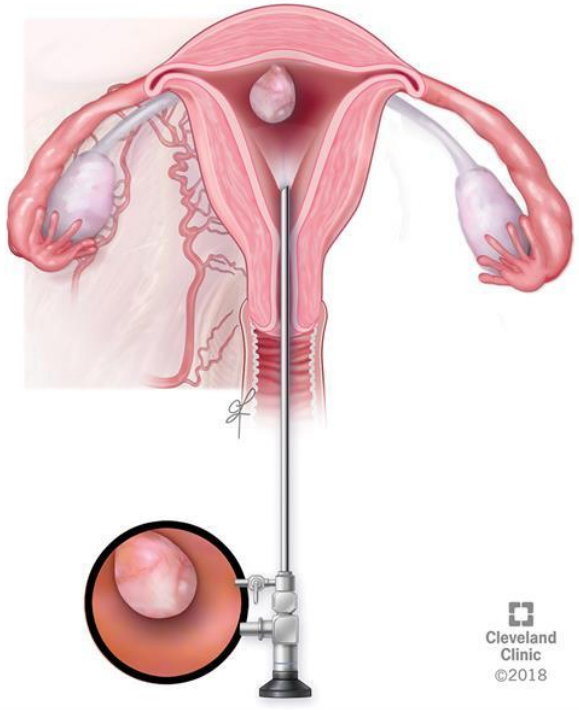


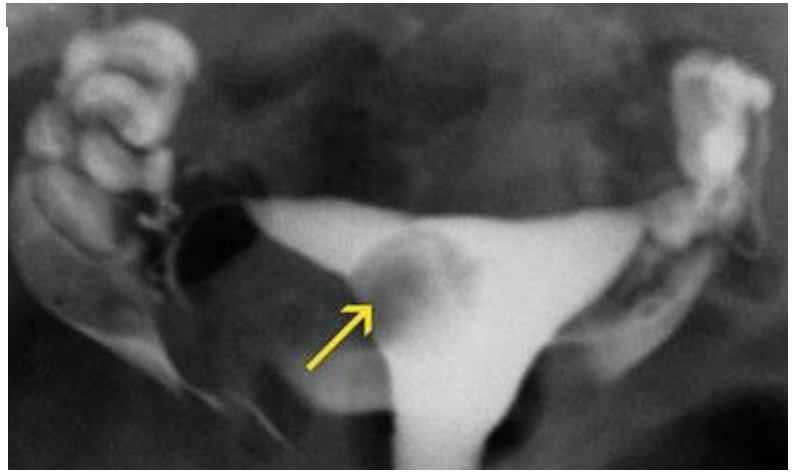
Endometrial Polyp: Evaluation and Diagnosis

Imaging
- Ultrasound (TVUS)
- HSG
- Sonohysterogram

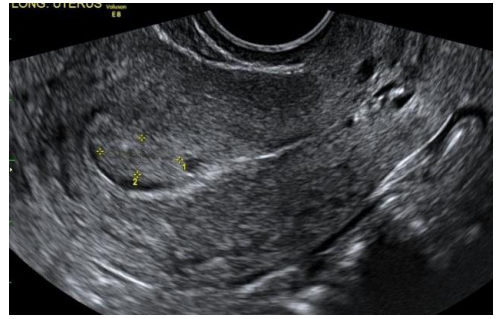
Hysteroscopy

Tissue diagnosis
- Endometrial biopsy
- Dilation and curettage



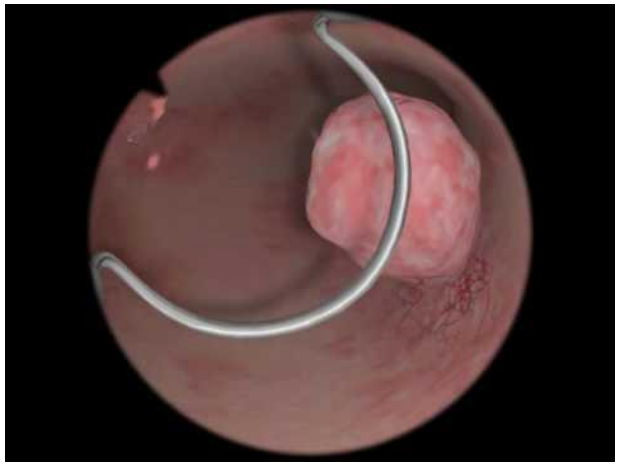






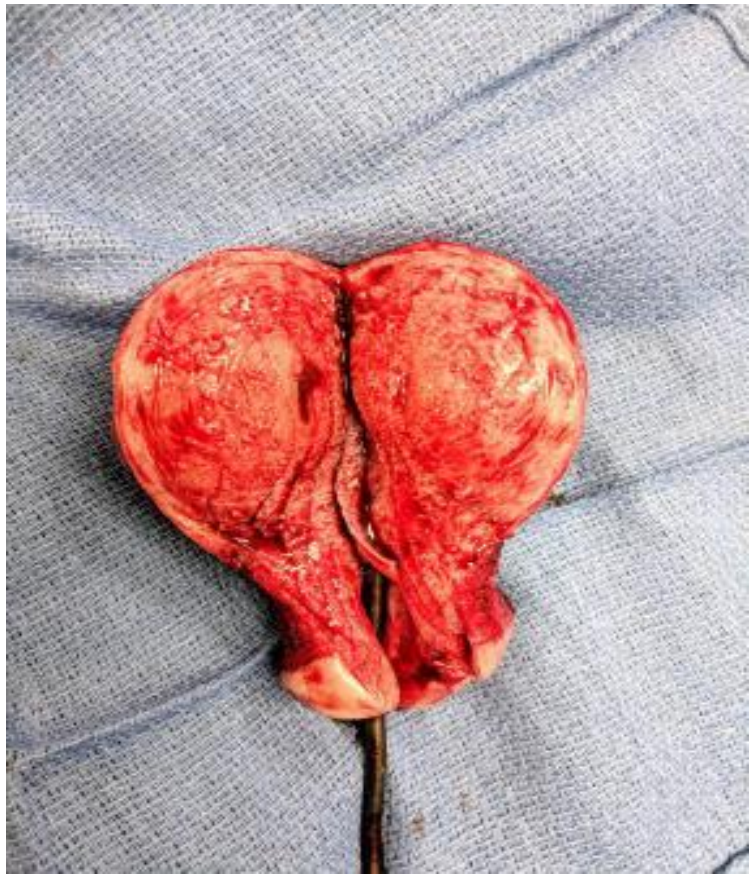
Endometrial Polyp: Treatment

- None necessary if asymptomatic
- Tissue removal
  - Hysteroscopic excision
  - Polypectomy
  - Curettage if small
- Pathology





PALM-COEIN



Adenomyosis

Endometrial glands and stroma present within the myometrium ("ectopic")

Cyclic thickening, breaking down and bleeding as within normally located endometrium

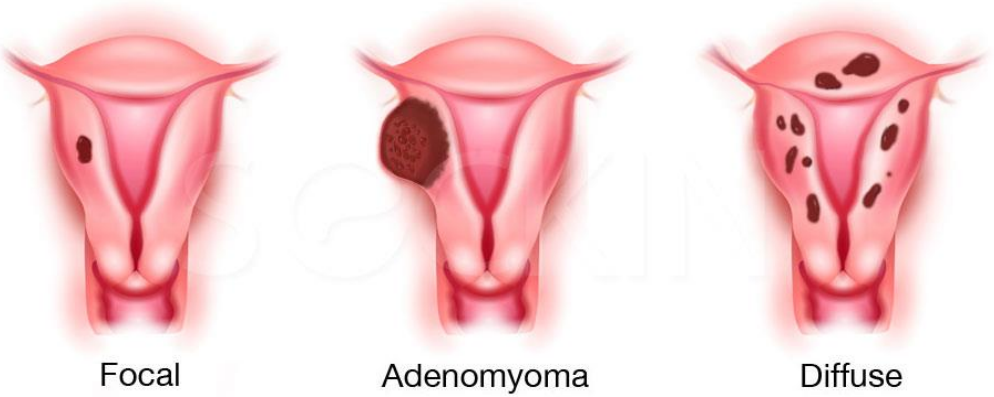
Resultant hypertrophy of surrounding myometrium

Cause?

- Cell migration due to disruption on endometrial-myometrial interface (D&C, uterine surgery, c/s)
- Metaplasia of pluripotent cells

Adenomyosis: Clinical Presentation

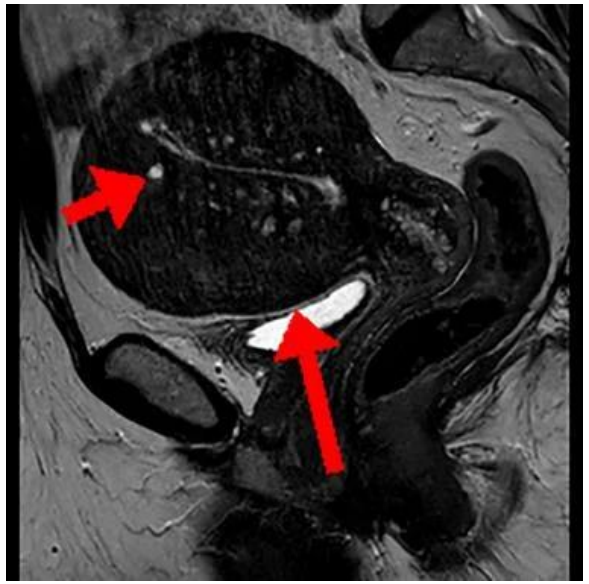
- Enlarged uterus
- Heavy menses
- Painful menses (dysmenorrhea)
- Dyspareunia
- Chronic pelvic pain
- Asymptomatic
- Infertility?



© Seckin Endometriosis Center for Medical Education and Research. All rights reserved.

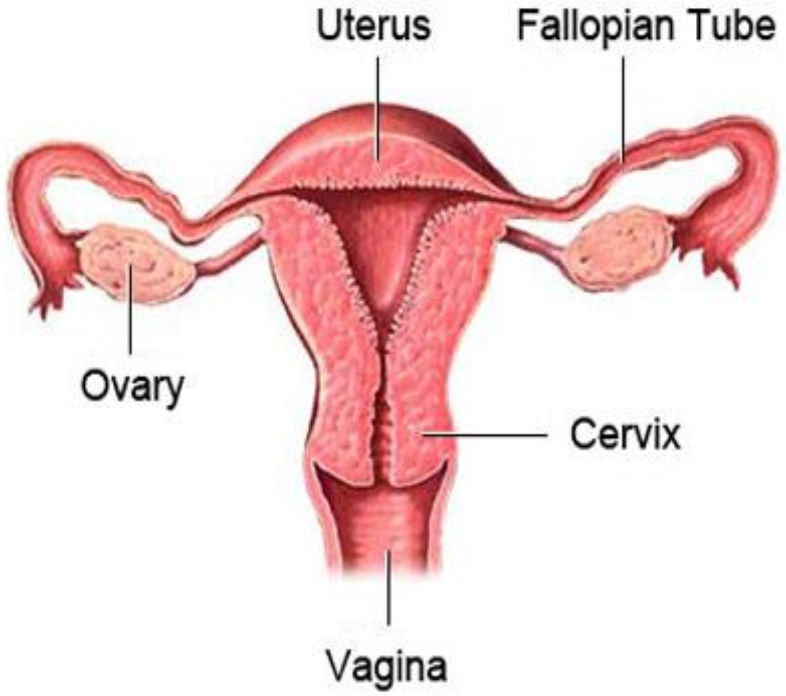
Adenomyosis: Evaluation and Diagnosis

- Exam- mobile, diffusely enlarged, "globular", soft
- Ultrasound
- MRI
- Surgical pathology- only definitive diagnosis
  - Endometrial glands and stroma present in myometrium

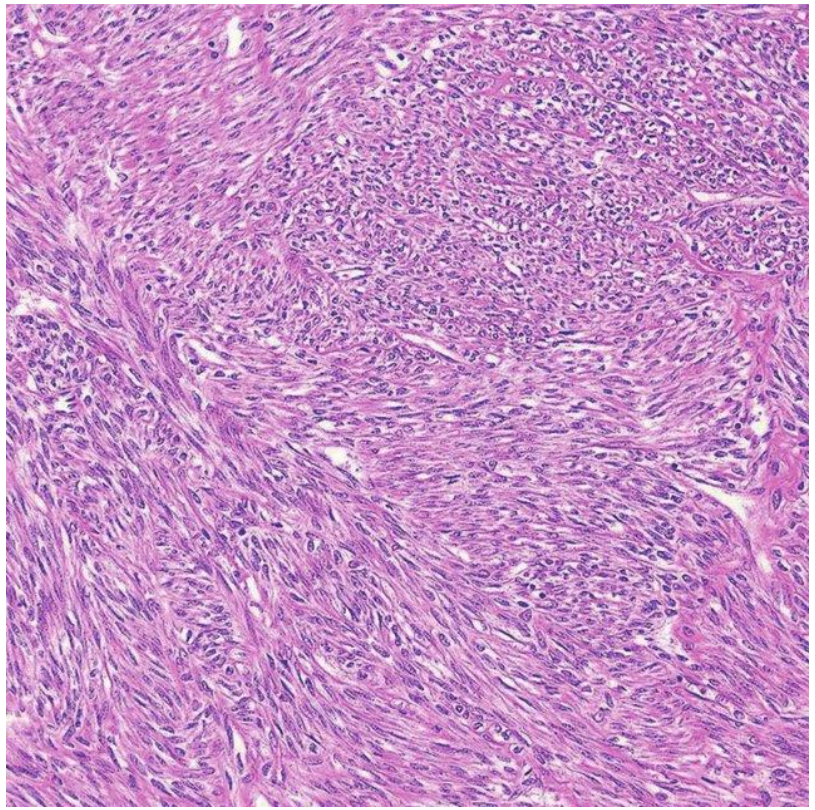


Adenomyosis: Treatment

- Hysterectomy
- Medical management of symptoms- Not curative
  - NSAIDs
  - Ooral contraceptive pills (OCPs)
  - Oral progestins
  - Progesterone IUDs



PALM-COEIN



Leiomyoma

"Fibroids" or "Myomas"

Most common pelvic tumor in people with a uterus

- 70-80% by age 50y
- Prevalence different by race
- Genetic predisposition within families

Benign monoclonal smooth muscle tumors (myometrial)- arises from a single cell

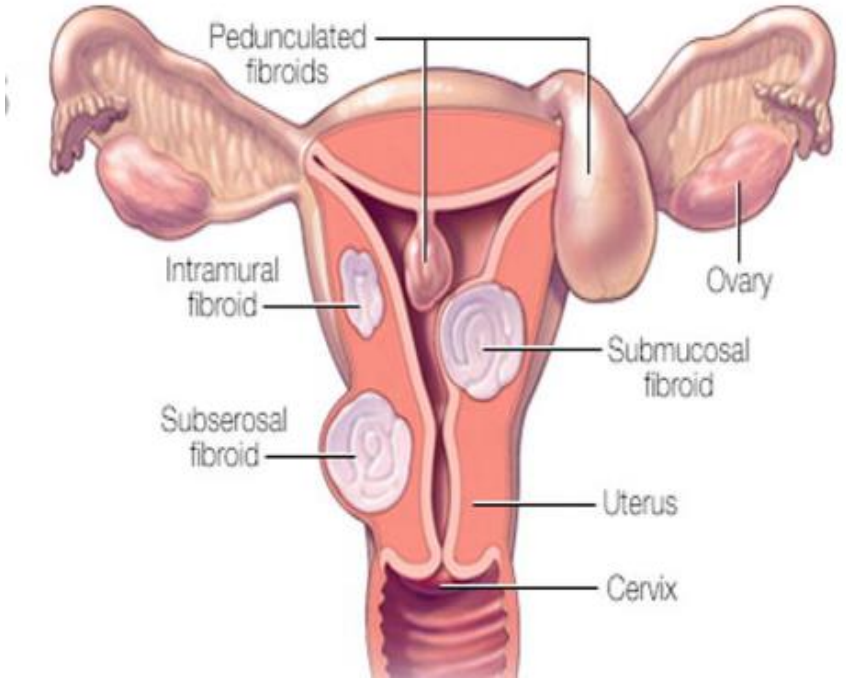
- Each myoma is from its own cell, NOT metastatic
- Somatic mutations often identified (translocations, duplications, deletions)
- Higher concentration of estrogen receptors than "normal" myometrium- hormone responsive

Variable location within uterine wall

Leiomyoma: Clinical Presentation

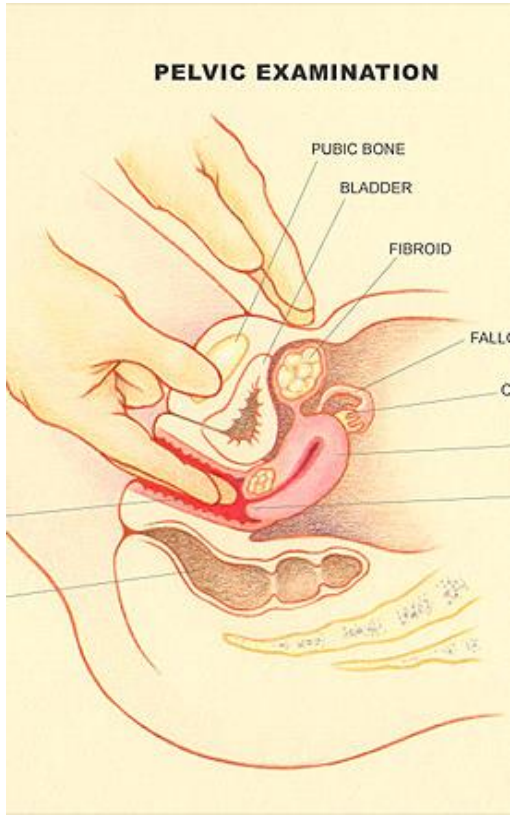
- Enlarged uterus/pelvic mass
- Heavy menses
- Pelvic pain
  - Dysmenorrhea
  - Dyspareunia
- "Bulk symptoms"/pressure

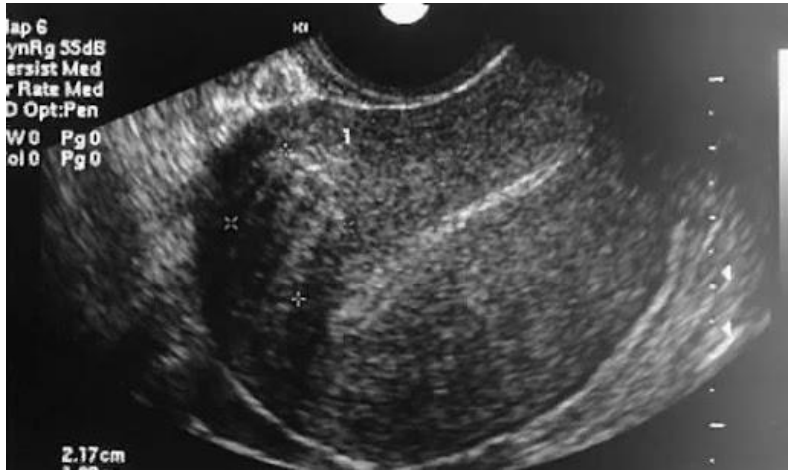
- Symptoms variable depending on location
- Rapidly growing or menopausal growth-
  - Uterine Sarcoma?

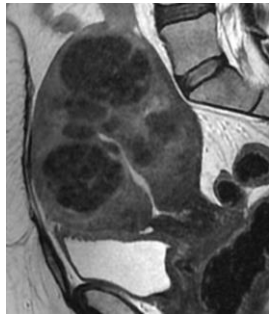


Leiomyoma: Evaluation and Diagnosis

- Exam- enlarged, irregular
- Ultrasound
- MRI







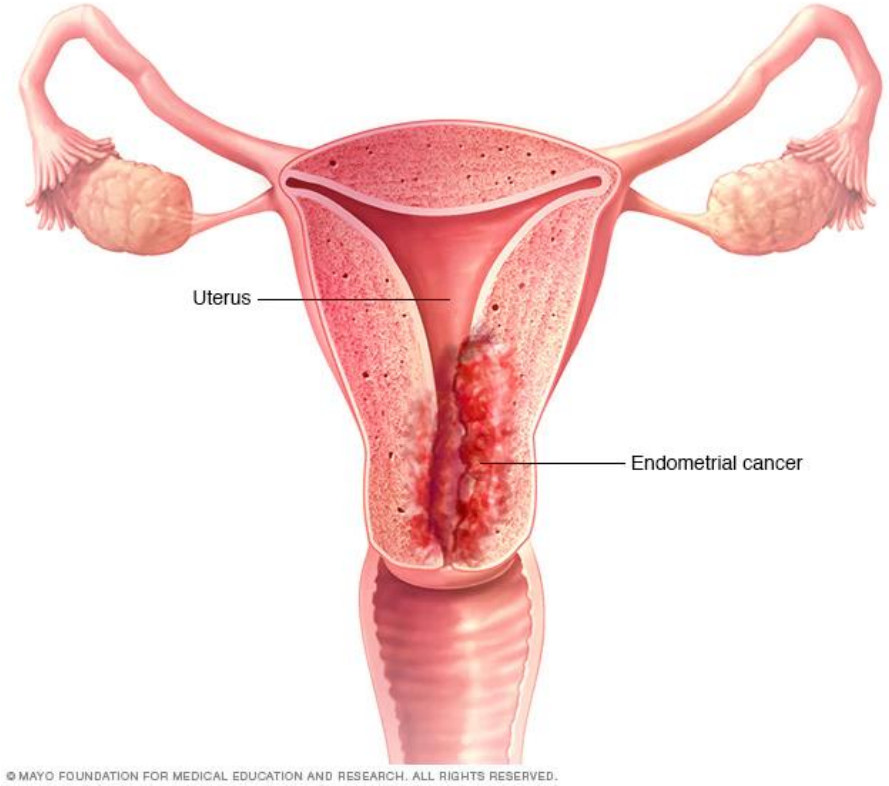


Leiomyoma: Treatment

- Hysterectomy
- Myomectomy
- Uterine artery embolization
- Medical management for symptoms and volume decrease



PAL**M**-COEIN



© MAYO FOUNDATION FOR MEDICAL EDUCATION AND RESEARCH. ALL RIGHTS RESERVED.

Malignancy & Hyperplasia

Endometrial cancer or hyperplasia

- Endometrial hyperplasia with cytological atypia
- Endometrial carcinoma
- Endometrial stromal sarcoma

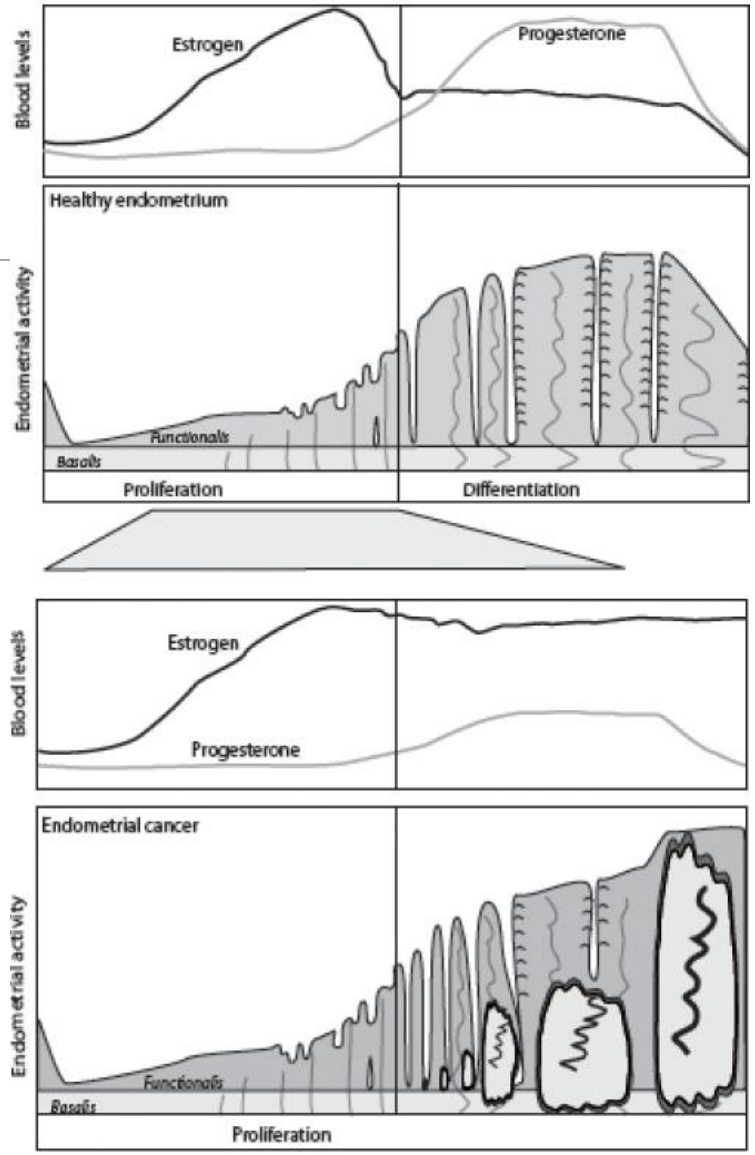
Less common to have malignancy in the premenopausal period

- Can happen with prolonged unopposed estrogen
- Must be considered given diagnostic/clinical importance

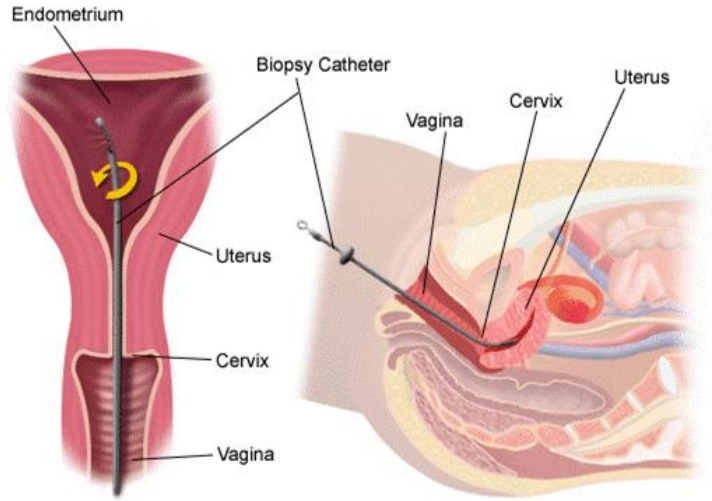
**Subclassification of endometrial hyperplasia and neoplasia using WHO or FIGO systems- outside the scope of this lecture

Malignancy & Hyperplasia: Clinical Presentation

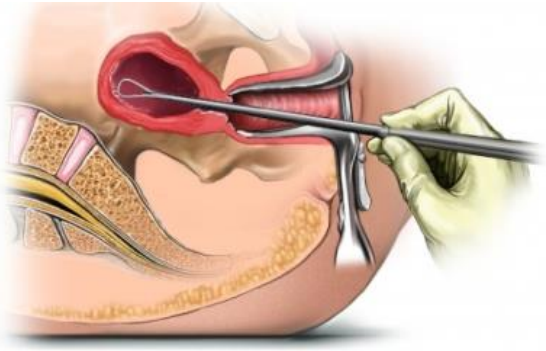
- Irregular, prolonged or heavy bleeding
- Prolonged anovulatory patients
- Postmenopausal



Malignancy & Hyperplasia: Evaluation and Diagnosis



- Endometrial tissue sampling
  - Endometrial biopsy (EMB)
  - Dilation and curettage (D&C)
  - Hysteroscopy guided biopsy

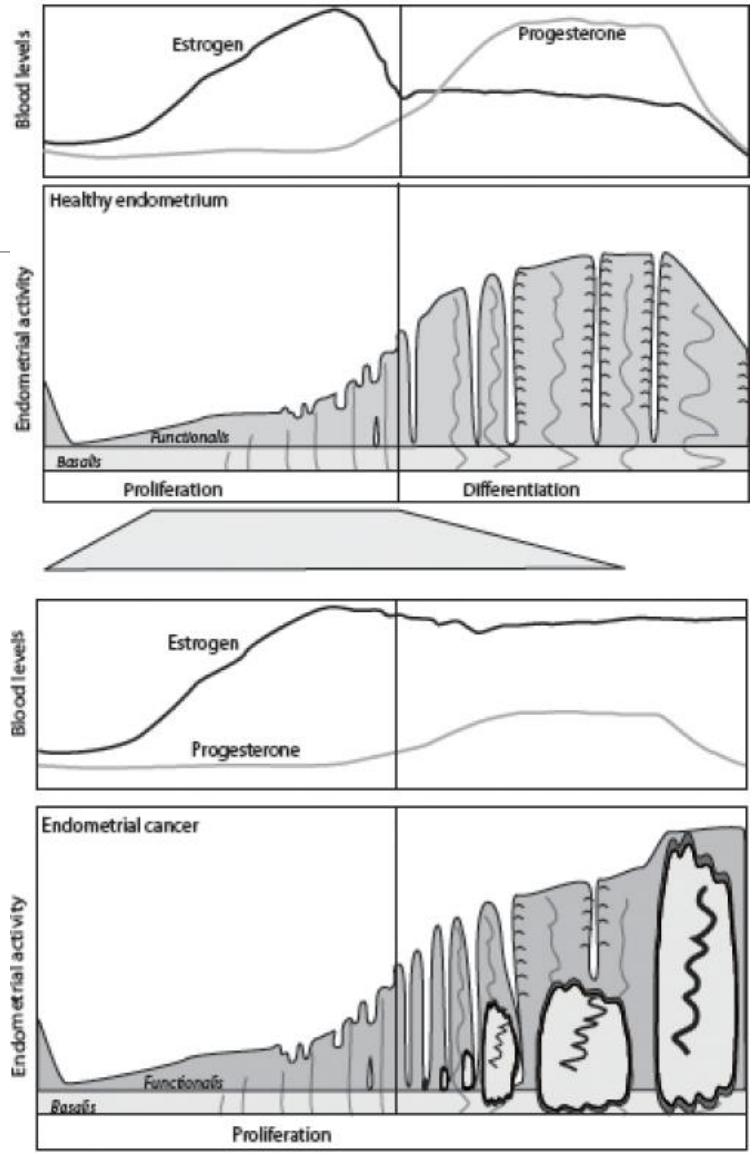




Malignancy & Hyperplasia: Treatment

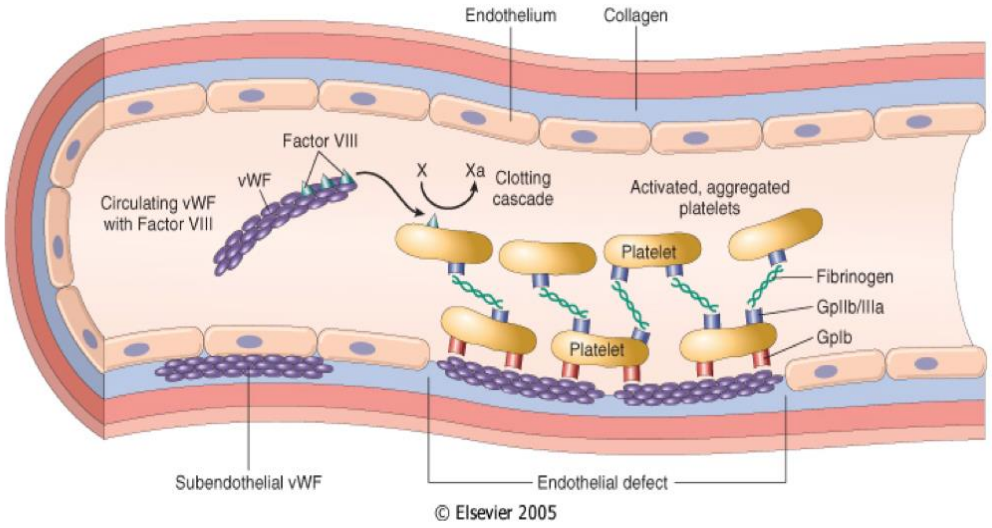
- Hysterectomy +/- lymphadenectomy
- Progesterone
  - Endometrial hyperplasia
  - Endometrial carcinoma in poor surgical candidate or fertility desired





Nonstructural Causes (COEIN)

PALM-COEIN



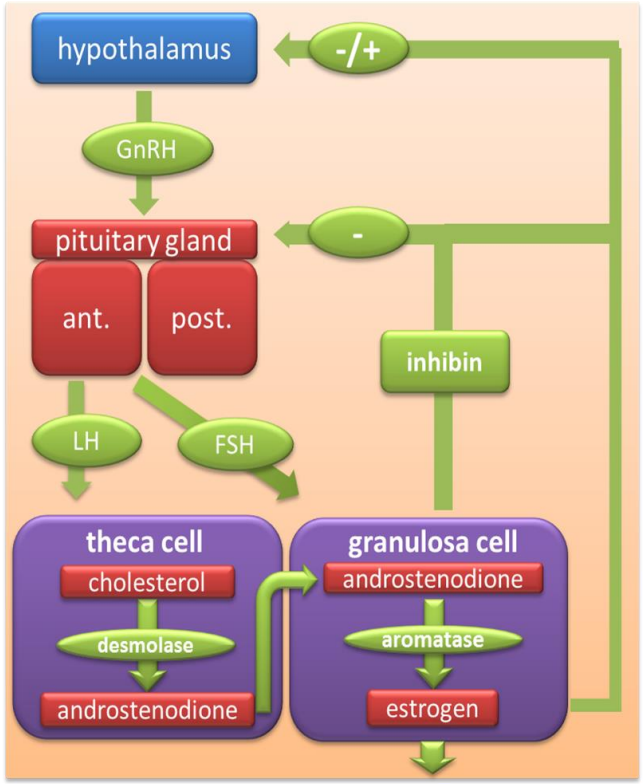
Coagulopathy

Inherited coagulopathy, most commonly von Willebrand disease
- Acquired coagulopathy less common
- Iatrogenic due to anticoagulation also included in this category

Diagnosis
- History
  - Heavy menses from menarche
  - Personal and family bleeding history
- Lab work
  - CBC
  - Coagulation panel (PT, aPTT, INR)
  - von Willebrand's testing

Treatment
- Hormonal therapy (OCP or progestin)
- Tranexamic acid (antifibrinolytic)
- Desmopressin?
  - At time of surgery/childbirth

PALM-cOEIN



Ovulatory Dysfunction

Spectrum: amenorrhea to irregular heavy periods

- Extremes of reproductive age
- Polyendocrine Metabolic Ovarian Syndrome (PMOS)

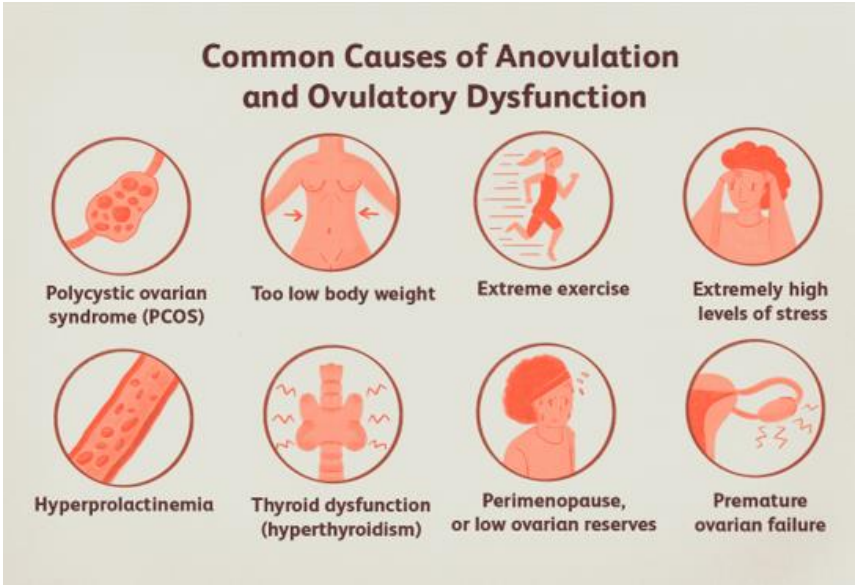
Disruption of hypothalamic-pituitary-ovarian axis

Mechanism of heavy menstrual bleeding (HMB) is unopposed estrogen causing marked thickening of the endometrium

Ovulatory Dysfunction: Causes

Physiologic
- Adolescence
- Perimenopause
- Lactation
- Pregnancy

Pathologic
- Hyperandrogenic (PMOS, CAH, adrogenic tumors)
- Hypothalamic dysfunction (calorie restriction)
- Hyperprolactinemia
- Thyroid disease
- Primary pituitary disease
- Premature ovarian failure
- Iatrogenic (chemo/radiation)
- Medications



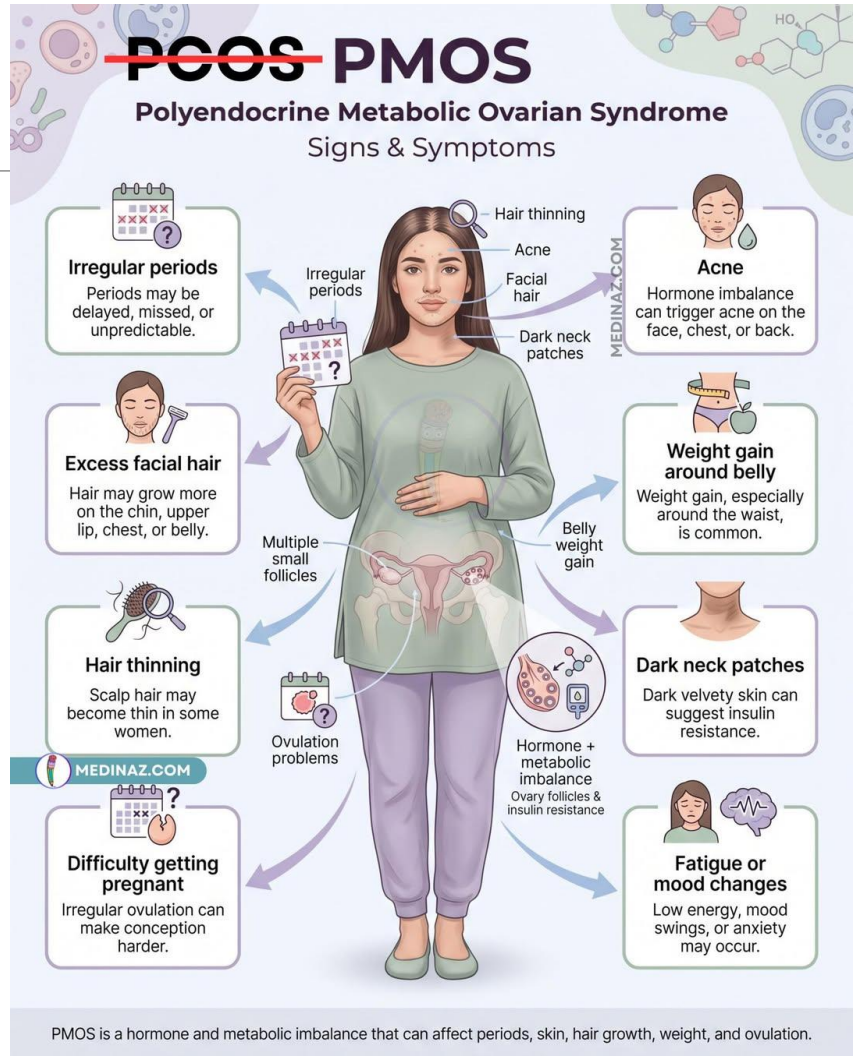
Ovulatory Dysfunction: Evaluation and Diagnosis

History and physical exam very important

Laboratory testing

- Beta HCG
- Prolactin
- Thyroid Stimulating Hormone (TSH)
- FSH
- Free testosterone
- Sex hormone binding globulin
- Estradiol
- Progesterone
- 17-OHP

- Consider endometrial sampling



Ovulatory Dysfunction: Treatment

Depends on identified cause and fertility/bleeding goals

- Treatment of underlying medical issue

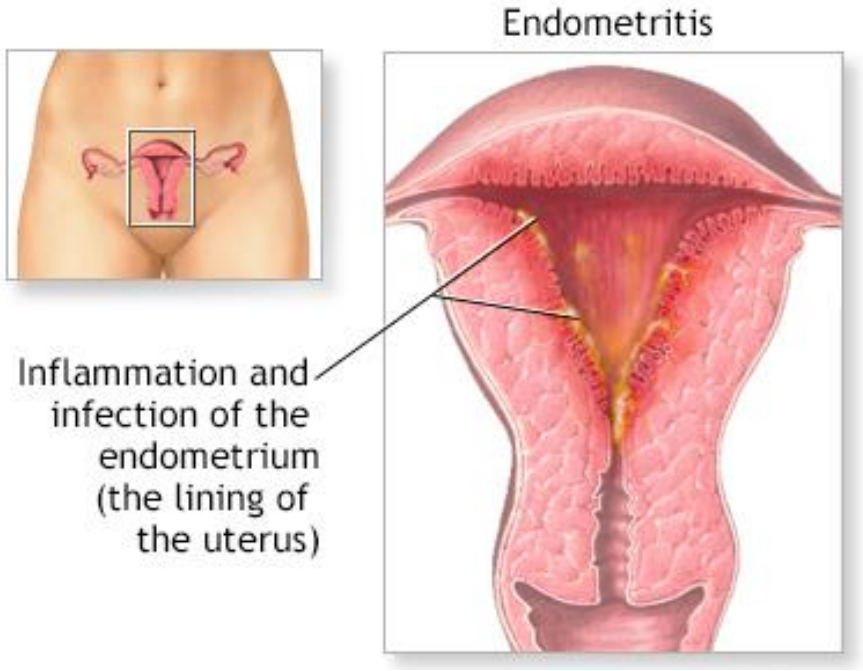
PMOS

- Often OCP, progesterone IUD or cyclic progesterone
- Weight loss
- Insulin sensitizing medications such as metformin
- Is fertility currently desired?
  - Withdrawal bleed followed by ovulation induction
  - First consider medical concerns in pregnancy





PALM-COEIN



Endometrial

HMB or IMB with structurally normal uterus

No evidence of coagulopathy

Thought to be a primary disorder of the endometrium

- Disorder of local hemostatic mechanisms
  - Deficiency in local vasoconstrictors
  - Local excess of plasminogen activator
  - Increased local vasodilator production
- Endometritis

No available tests or biomarkers

Diagnosis based on exclusion of other causes and history

Endometrial Dysfunction: Treatment

OCPs

Tranexamic acid

Antibiotics- Doxycycline?

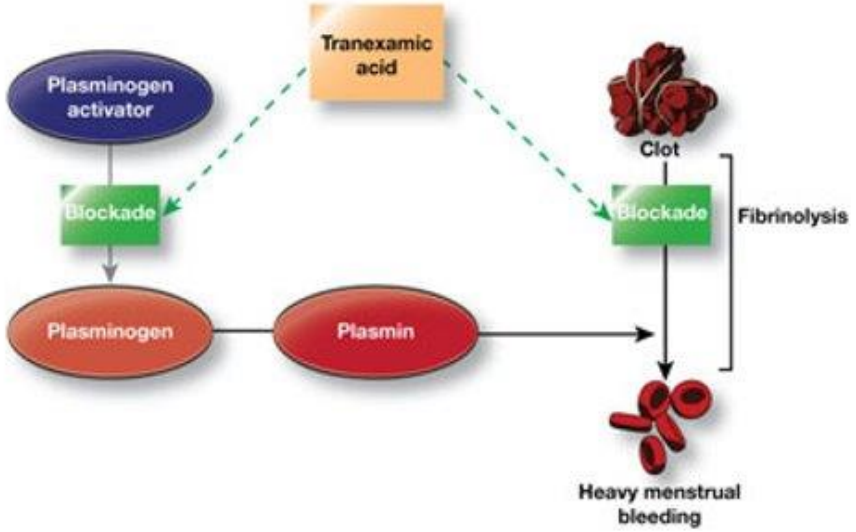
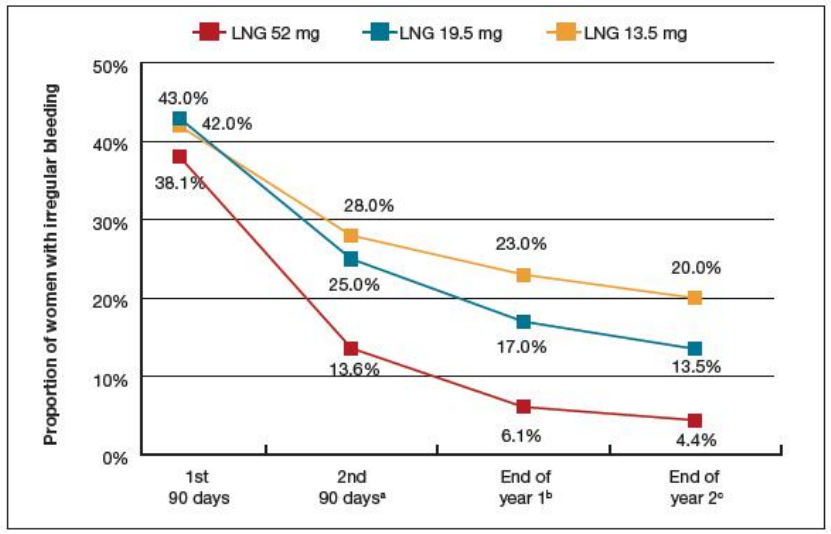


FIGURE How tranexamic acid stanches heavy menstrual bleeding

PALM-COEIN

FIGURE 2 Irregular bleeding rates for users of the 52-mg, 19.5-mg, and 13.5-mg LNG IUD



[a] 52 mg vs 19.5 mg: RR, 0.54 (95% CI, 0.47–0.63)
52 mg vs 13.5 mg: RR, 0.48 (95% CI, 0.42–0.56)
19.5 mg vs 13.5 mg: RR, 0.89 (95% CI, 0.79–1.00)
[b] 52 mg vs 19.5 mg: RR 0.37 (95% CI, 0.30–0.47)
52 mg vs 13.5 mg: RR, 0.27 (95% CI, 0.21–0.34)
19.5 mg vs 13.5 mg: RR, 0.73 (95% CI, 0.63–0.85)
[c] Year 2 estimated by interpolation (data available only for year 1 and year 3). Therefore, statistical testing at end-of-year 2 not calculated.
Abbreviations: CI, confidence interval; IUD, intrauterine device; LNG, levonorgestrel; RR, relative risk.
Adapted from Goldthwaite LM, Creinin MD. Comparing bleeding patterns for the levonorgestrel 52 mg, 19.5 mg, and 13.5 mg intrauterine systems. *Contraception*. 2019;100:128-131.

Iatrogenic

Hormonal medications- will often resolve over time
- Combined hormonal contraceptives
- Progesterone only contraceptives
  - Depo-provera
  - Etonogestrel implant
  - Progesterone IUDs

Copper IUD



PALM-COEI**N**



Not Yet Classified

AV malformations?

Myometrial hypertrophy?

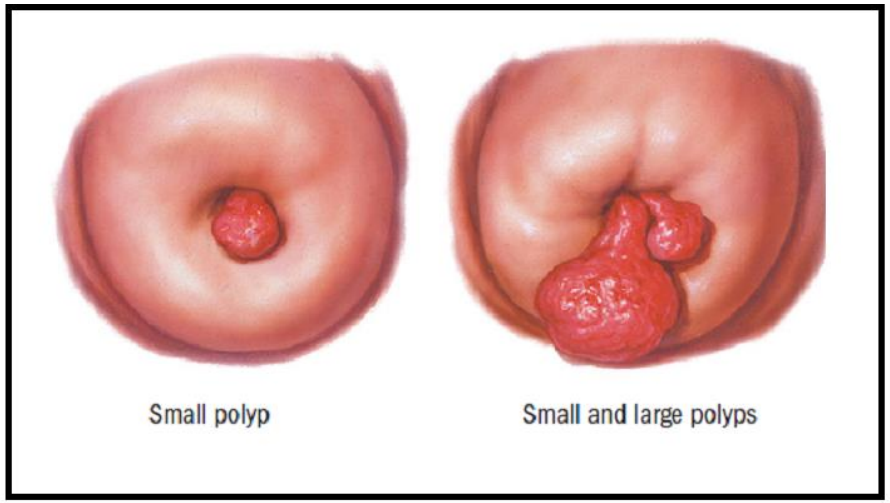
Chronic endometritis?

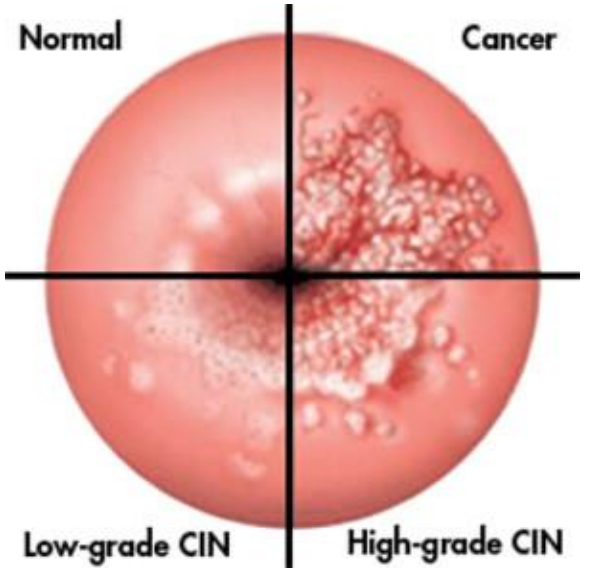
"Future entities"

Other Vaginal Bleeding

Non-uterine Bleeding

- Vulva, vagina, cervix, rectum, urethra/bladder
- Need full pelvic exam (speculum and bimanual)
  - Rectal exam or referral if no gyn cause of bleeding identified
- Consider pap and biopsy of any visible lesions





Postmenopausal Bleeding

All bleeding after menopause is abnormal and needs evaluation

- Atrophy - endometrial or vaginal
- Endometrial hyperplasia/neoplasia
- Polyp- cervical or vaginal

Imaging Overview

Ultrasound- both abdominal and transvaginal

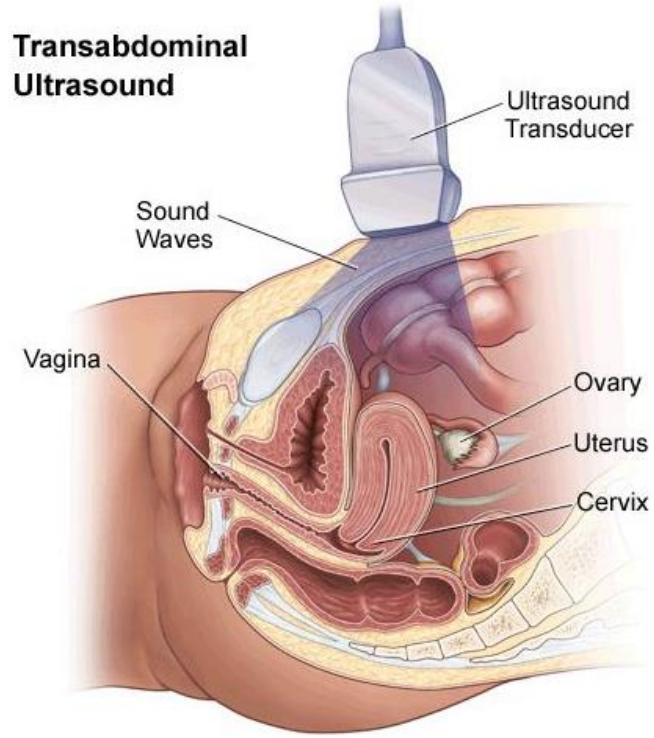
Sonohysterogram

Hysterosalpingogram

Hysteroscopy

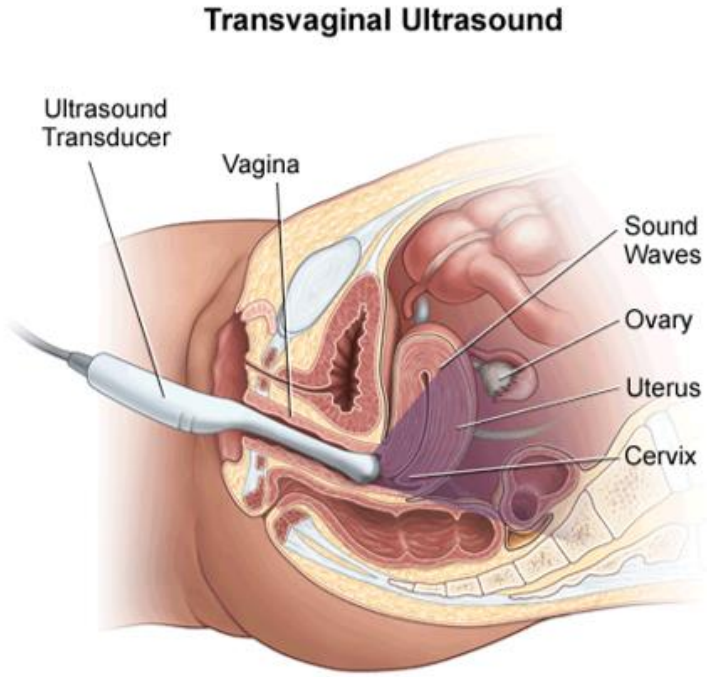
MRI

CT not great for defining/characterizing pelvic masses



Ultrasound

High frequency sound waves create images



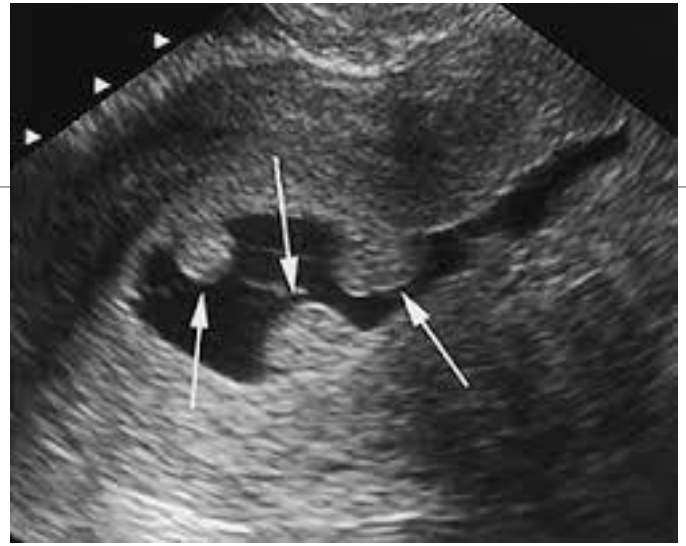


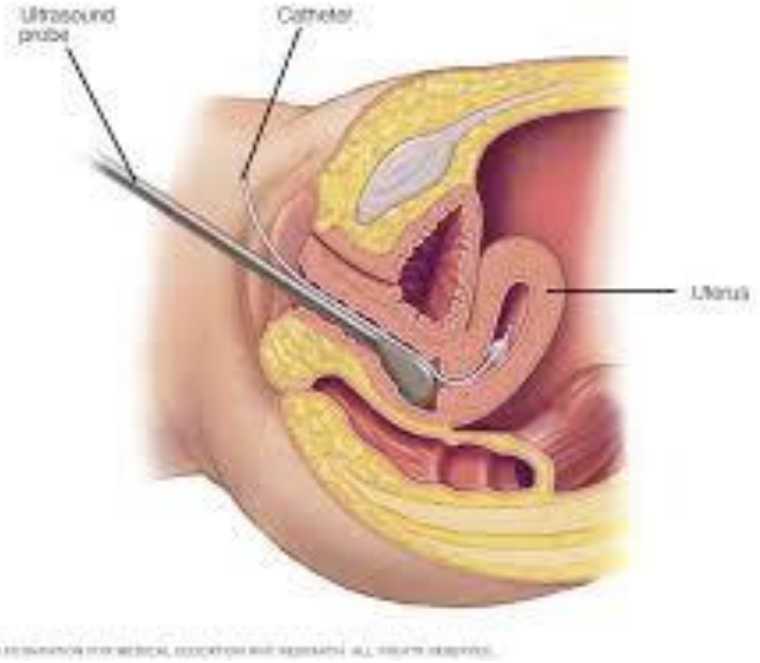


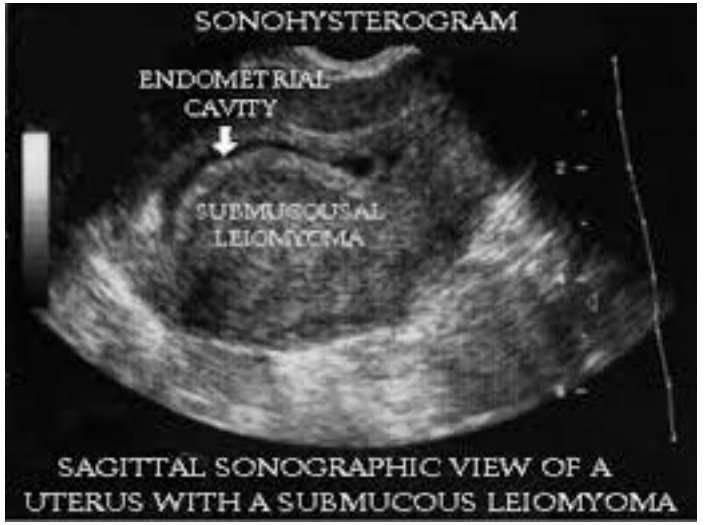


Sonohysterogram

Inject saline through cervix and use ultrasound to visualize intracavitary lesions







Hysterosalpingogram



Catheter placed through cervix

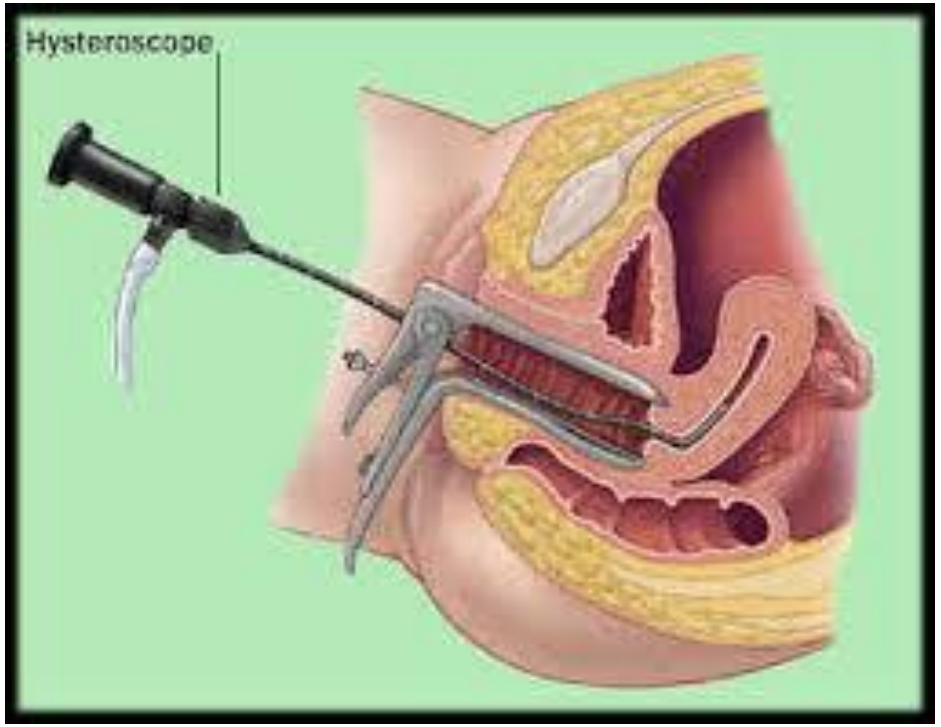
Contrast introduced into uterus

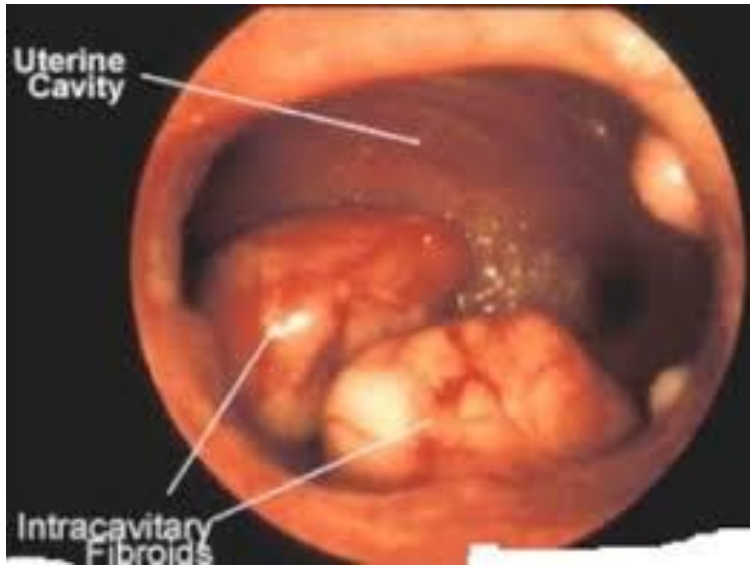
Xray (fluoroscopy) shows the contrast filling the uterus and uterine tubes and spilling out the ends of the tubes

Can identify intracavitary structural lesions (as well as tubal disease)

Hysteroscopy

Camera through cervix to directly visualize (and treat, with operative hysteroscopy) intracavitary lesions

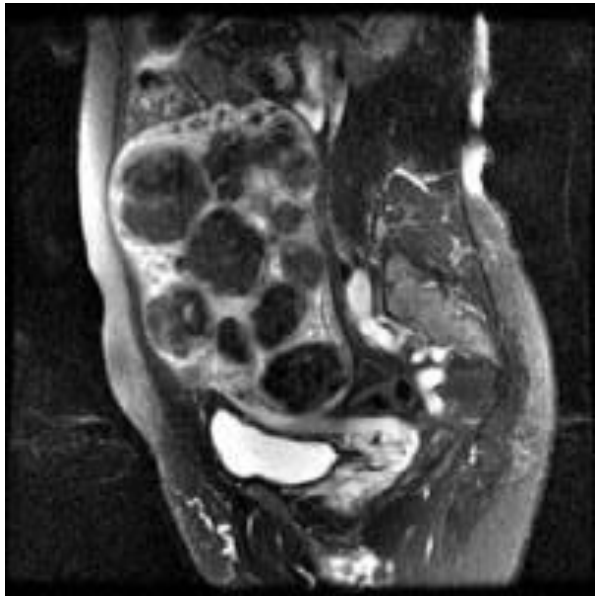




MRI

Used for diagnosis and surgical planning, especially for robotic surgery





Patient Goals of Treatment

Reassurance vs treatment

Contraception needs

Fertility desires

Acceptable bleeding pattern

Ability to comply with dosing

Definitive treatment vs quick return to work

Chronic Pelvic Pain

Chronic Pelvic Pain Defined

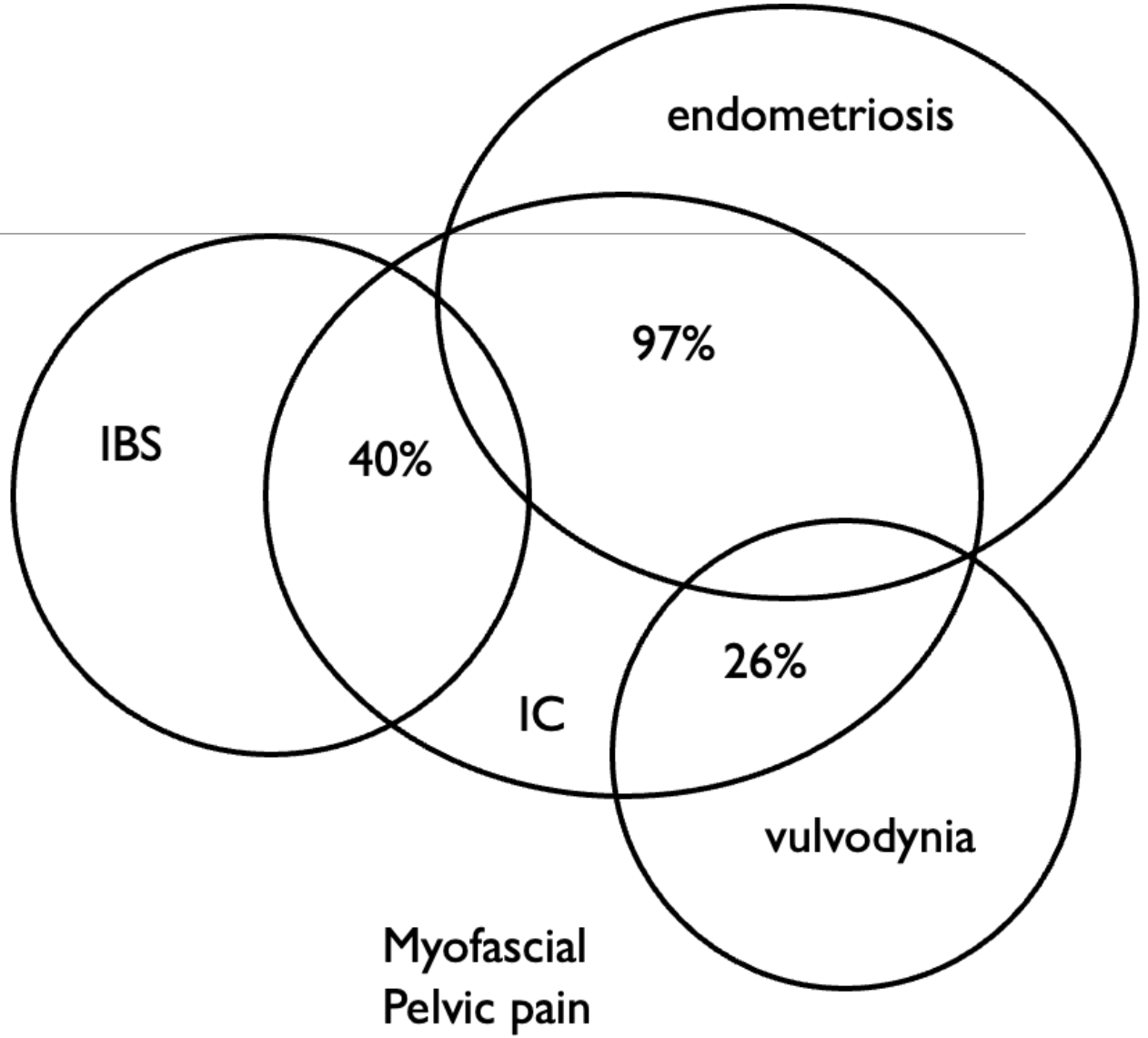
- Six month's duration of pain below umbilicus, in the abdomen, pelvic, groin, or lower back
- Severe enough to cause functional disability

Differential Diagnosis

- GYN- Endometriosis, PID, pelvic congestion, leiomyoma, adenomyosis, malignancy, vulvodynia, vestibulitis, adhesive disease, pelvic organ prolapse
- Uro- Interstitial cystitis/painful bladder syndrome, recurrent UTI, radiation cystitis, malignancy
- GI- IBS, IBD, diverticulitis, chronic constipation, celiac, malignancy
- MSK- pelvic floor pain, DJD, pelvic girdle pain, labral tear, arthritis
- Neuro- nerve entrapment, peripheral neuropathy
- Psych- Somatization, physical/sexual abuse, depression

There is a lot of overlap between many of these chronic conditions

Treatment can be difficult and often needs to be multimodal



Common Gynecologic Causes

Leiomyoma

Adenomyosis

Pelvic adhesive disease

Pelvic Inflammatory Disease (PID)

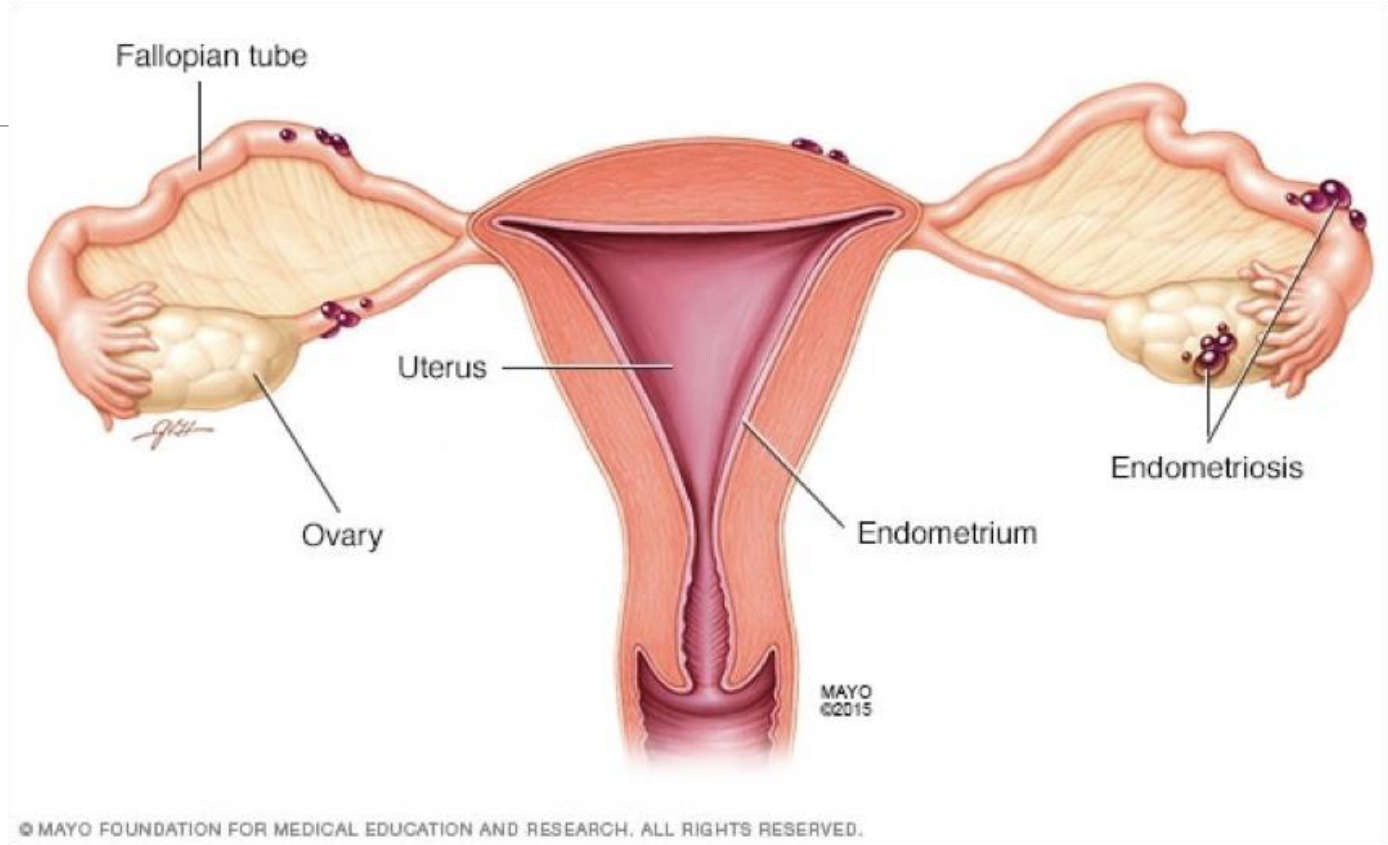
Endometriosis

- Endometrial glands and stroma outside of the uterus
- Chronic inflammatory condition leading to scarring

Endometriosis

Most common sites are

- Ovaries 65%
- Posterior cul-de-sac 30%
- Uterosacral ligaments 45%
- Fallopian tubes
- Appendix
- Round ligament



- Can also be seen in distal sites like the lungs and brain

Endometriosis Theories

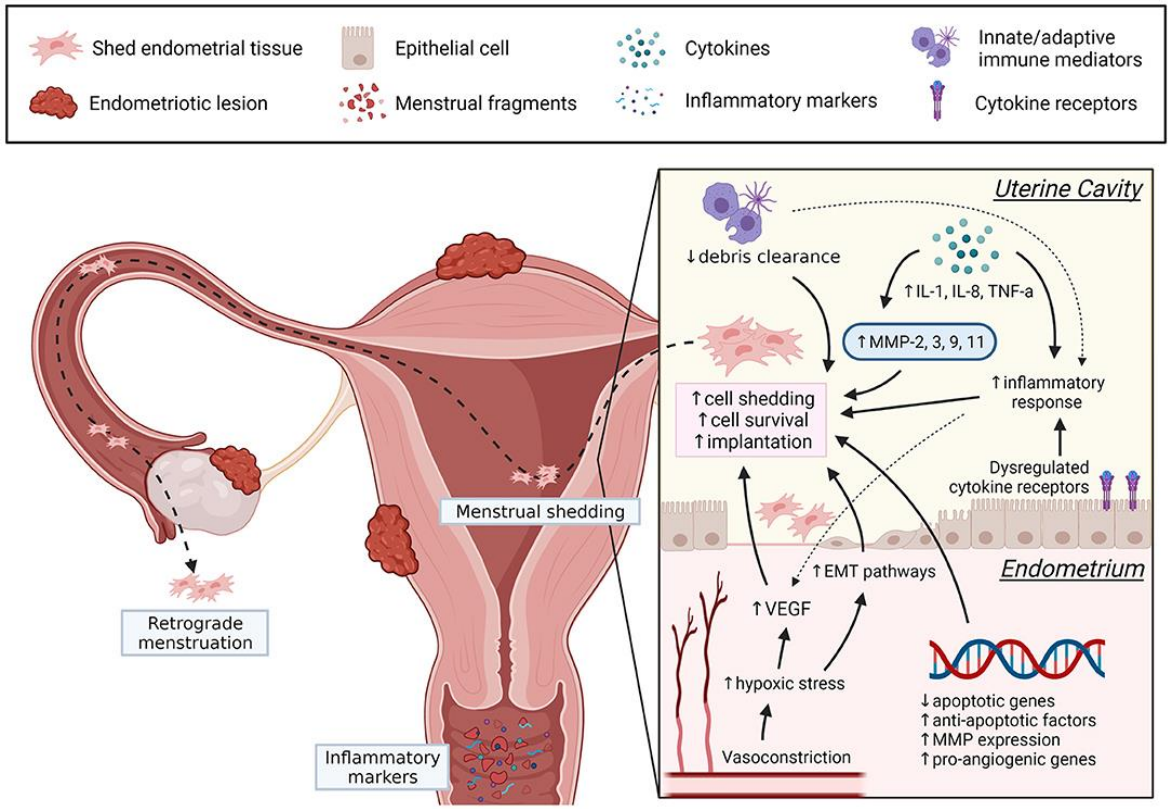
Retrograde menstruation
- With subsequent implantation into the peritoneal cavity

Coelomic metaplasia

Hematogenous/Lymphatic spread

Stem cell recruitment theory
- Differentiation of hematopoietic stem cells at different anatomical sites

Embryogenetic theory
- Residual Mullerian duct cells develop into endometriosis lesions with estrogen exposure



Diagnosis of Endometriosis

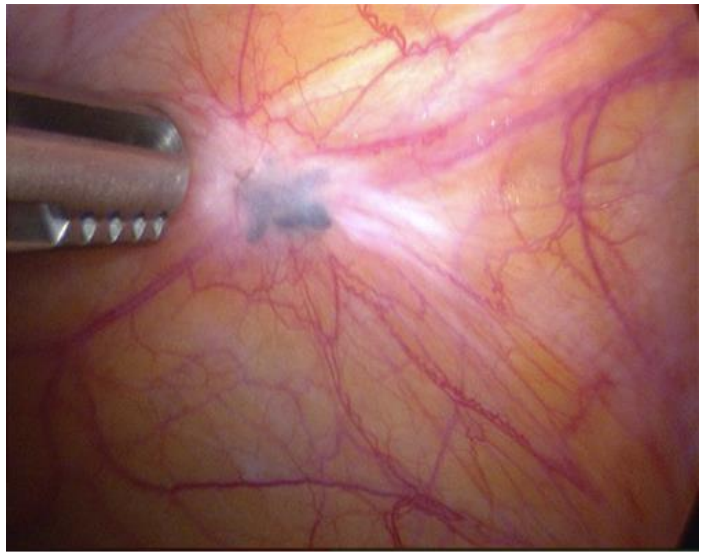
Clinical presentation

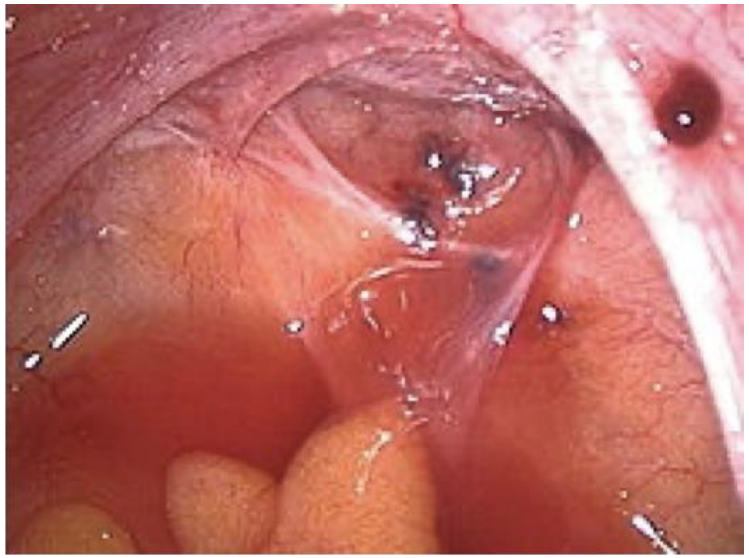
- Pelvic pain
  - Dysmenorrhea
  - Dyspareunia
  - Back pain, abdominal pain
  - Pain with urination or BMs

Exam

- Uterosacral nodularity
- Immobile cervix/uterus
- Pain with elevation of uterus

Endometriosis on laparoscopy







Endometriosis Treatment

Medical Therapy

- NSAIDS
- OCPs are first line
- Progestin therapy
- GnRH Agonist (Leuprolide)
- GnRH Antagonist (Elagolix)- effective immediately, no initial LH surge

Surgery

References

Diagnosis of Abnormal Uterine Bleeding in Reproductive Age Women ACOG Practice Bulletin Number 128, July 2012

Management of Abnormal Uterine Bleeding Associated with Ovulatory Dysfunction ACOG Practice Bulletin 136, July 2013

FIGO classification system (PALM-COEIN) for causes of abnormal uterine bleeding in nongravid women of reproductive age, International Journal of Obstetrics and Gynecology 113, Jan 2011